

15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10

Estimation Results for Exercise 15.6

	(1) OLS 1987	(2) OLS 1988	(3) FE	(4) FE Robust	(5) RE
<i>C</i>	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
<i>EXPER</i>	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
<i>EXPER</i> ²	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
<i>SOUTH</i>	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
<i>UNION</i>	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
<i>N</i>	716	716	1432	1432	1432

(standard errors in parentheses)

- a. The OLS estimates of the $\ln(\text{WAGE})$ model for each of the years 1987 and 1988 are reported in columns (1) and (2). How do the results compare? For these individual year estimations, what are you assuming about the regression parameter values across individuals (heterogeneity)?
- b. The $\ln(\text{WAGE})$ equation specified as a panel data regression model is

$$\ln(\text{WAGE}_{it}) = \beta_1 + \beta_2 \text{EXPER}_{it} + \beta_3 \text{EXPER}_{it}^2 + \beta_4 \text{SOUTH}_{it} + \beta_5 \text{UNION}_{it} + (u_i + e_{it}) \quad (\text{XR15.6})$$

Explain any differences in assumptions between this model and the models in part (a).

- c. Column (3) contains the estimated fixed effects model specified in part (b). Compare these estimates with the OLS estimates. Which coefficients, apart from the intercepts, show the most difference?
- d. The F -statistic for the null hypothesis that there are no individual differences, equation (15.20), is 11.68. What are the degrees of freedom of the F -distribution if the null hypothesis (15.19) is true? What is the 1% level of significance critical value for the test? What do you conclude about the null hypothesis?
- e. Column (4) contains the fixed effects estimates with cluster-robust standard errors. In the context of this sample, explain the different assumptions you are making when you estimate with and without cluster-robust standard errors. Compare the standard errors with those in column (3). Which ones are substantially different? Are the robust ones larger or smaller?
- f. Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

(a)

二年的估計係數非常接近，EXPER在1987為0.1270，在1988為0.1265
SOUTH在1987為-0.2128，1988為-0.2384。SE也很接近。
表示這二年決定女性薪金的結構性參數變化不大。

迴歸分析的假設：

1. 參數對所有個體皆相同

2. 未觀察到的異質性與解釋變數不相關

違反假設，OLS估計產生系統變數偏誤。

(b)

1. 誤差項分解 $u_i + e_{it}$ ，其中 u_i 為個體固定效果，代表
不隨時間改變的個人特徵，OLS忽略此項，Panel data
regression 有考慮

2. OLS 不滿足一致的假設： $\text{Cov}(X_i, e_i) \neq 0$

Panel data:

① $\text{Cov}(X_{it}, u_i) \neq 0 \Rightarrow \text{Fixed Effects}$

② $\text{Cov}(X_{it}, u_i) = 0 \Rightarrow \text{Random Effects}$

1b) 3. OLS: 716 個樣本彼此之間獨立 (iid)

Panel data: 同一個體在不同時間的誤差存在
serial correlation / autocorrelation.

4. 同時使用二期的資料

1c) EXPER 和 SOUTH 差異最大

1. EXPER: OLS 係 0.1270, FE 係 0.1265

因為 OLS 假設 $\text{cov}(x_i, e_i) = 0$, 把個人能力... 歸因於
EXPER, 導致係數高估, 產生正的遺漏變量偏誤.

Fixed effect model 排除 x_i 的影響, 得到的係數
較接近真實狀況。

2. SOUTH: OLS 係 -0.1128, FE 係 -0.1261

控制個體固定效果後, 同一個人從非南方搬到南方,
其薪資的負面衝擊, 比橫斷面比較還要大。

(d)

$$F = \frac{(SS E_R - SS E_U) / (N-1)}{SS E_U / (NT - N - k_s)} = 11.68$$

$$N = 716, T = 2, k_s = 4$$

$$H_0: \beta_{11} = \beta_{12}, \beta_{12} = \beta_{13}, \dots, \beta_{1,N-1} = \beta_{1N}$$

$$H_1: \text{the } \beta_{1i} \text{ are not all equal}$$

$$\text{分子 } df = 716 - 1 = 715$$

$$\text{分母 } df = 716 \times 2 - 716 - 4 = 712$$

$$F_{0.01}(715, 712) = 1.1905 \quad (\text{By R})$$

$$F^* = 11.68 > F_{0.01}(715, 712)$$

∴ reject H_0 . 有證據支持個體間存在顯著且無法被觀察到的個體差異. 因此, 使用OLS會產生偏誤, 使用 Fixed effect model 較合適。

if)

1. EXPER: FE 偏 0.0575, RE 偏 0.0986.

RE 係數較高, 因為 RE 假設 $\text{cov}(x_{it}, u_i) = 0$,
會把個人能力歸到 EXPER, 導致 RE 係數高估

2. SOUTH: FE 偏 -0.3261, RE 偏 -0.2326

$$t = \frac{b_{FE,k} - b_{RE,k}}{\left[\widehat{\text{var}}(b_{FE,k}) - \widehat{\text{var}}(b_{RE,k}) \right]^{1/2}} = \frac{b_{FE,k} - b_{RE,k}}{\left[\text{se}(b_{FE,k})^2 - \text{se}(b_{RE,k})^2 \right]^{1/2}}$$

H_0 : u_i 與所有解釋變數無關

$$t^x = 15.36 > t^*_{0.99}(14) = 9.488$$

在 5% 的顯著水準下, 拒絕 H_0 , 有證據反指

u_i 和解釋變數有相關, 所以應使用 FE 而非
RE 來進行估計。

- 15.20** This exercise uses data from the STAR experiment introduced to illustrate fixed and random effects for grouped data. In the STAR experiment, children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star*.
- Estimate a regression equation (with no fixed or random effects) where *READSCORE* is related to *SMALL*, *AIDE*, *TCHEXPER*, *BOY*, *WHITE_ASIAN*, and *FREELUNCH*. Discuss the results. Do students perform better in reading when they are in small classes? Does a teacher's aide improve scores? Do the students of more experienced teachers score higher on reading tests? Does the student's sex or race make a difference?
 - Reestimate the model in part (a) with school fixed effects. Compare the results with those in part (a). Have any of your conclusions changed? [*Hint*: specify *SCHID* as the cross-section identifier and *ID* as the "time" identifier.]
 - Test for the significance of the school fixed effects. Under what conditions would we expect the inclusion of significant fixed effects to have little influence on the coefficient estimates of the remaining variables?
 - Reestimate the model in part (a) with school random effects. Compare the results with those from parts (a) and (b). Are there any variables in the equation that might be correlated with the school effects? Use the LM test for the presence of random effects.
 - Using the *t*-test statistic in equation (15.36) and a 5% significance level, test whether there are any significant differences between the fixed effects and random effects estimates of the coefficients on *SMALL*, *AIDE*, *TCHEXPER*, *WHITE_ASIAN*, and *FREELUNCH*. What are the implications of the test outcomes? What happens if we apply the test to the fixed and random effects estimates of the coefficient on *BOY*?
 - Create school-averages of the variables and carry out the Mundlak test for correlation between them and the unobserved heterogeneity.

(a)

```
Call:
lm(formula = readscore ~ small + aide + tchexper + boy + white_asian +
    freelunch, data = star)
```

Residuals:

Min	1Q	Median	3Q	Max
-107.220	-20.214	-3.935	14.339	185.956

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	437.76425	1.34622	325.180	< 2e-16 ***
small	5.82282	0.98933	5.886	4.19e-09 ***
aide	0.81784	0.95299	0.858	0.391
tchexper	0.49247	0.06956	7.080	1.61e-12 ***
boy	-6.15642	0.79613	-7.733	1.23e-14 ***
white_asian	3.90581	0.95361	4.096	4.26e-05 ***
freelunch	-14.77134	0.89025	-16.592	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 5759 degrees of freedom

(20 observations deleted due to missingness)

Multiple R-squared: 0.09685, Adjusted R-squared: 0.09591

F-statistic: 102.9 on 6 and 5759 DF, p-value: < 2.2e-16

$$\text{Readscore} = \beta_0 + \beta_1 \text{small} + \beta_2 \text{Aide} + \beta_3 \text{Tchexper} + \beta_4 \text{Boy} \\ + \beta_5 \text{White} - \text{asian} + \beta_6 \text{Freelunch} + e$$

small 係改正顯著, 代表小班的 Readsore 會較高

Aide 係改不顯著, 表示 Aide 對於 Readsore 的影響不明顯

Tchexper 係改正顯著, 表示教師經驗每增加一年, 學生的 Readsore 會提高.

Boy 係改負顯著, 代表男生的 Readsore 低於女生

White - asian 係改正顯著, 代表 White or Asian 的學生 Readsore 較高.

Freelunch 係改負顯著, 表示低收入的学生 Readsore 較低

(b)

Oneway (individual) effect Within Model

Call:

```
plm(formula = readscore ~ small + aide + tchexper + boy + white_asian +  
freelunch, data = pdata, effect = "individual", model = "within")
```

Unbalanced Panel: n = 79, T = 34-137, N = 5766

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-102.6381	-16.7834	-2.8473	12.7591	198.4169

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
small	6.490231	0.912962	7.1090	1.313e-12 ***
aide	0.996087	0.881693	1.1297	0.2586
tchexper	0.285567	0.070845	4.0309	5.629e-05 ***
boy	-5.455941	0.727589	-7.4987	7.440e-14 ***
white_asian	8.028019	1.535656	5.2277	1.777e-07 ***
freelunch	-14.593572	0.880006	-16.5835	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 4628000

Residual Sum of Squares: 4268900

R-Squared: 0.077592

Adj. R-Squared: 0.063954

F-statistic: 79.6471 on 6 and 5681 DF, p-value: < 2.22e-16

Small 係改變大, 表示小班制影响变大

Aide 結果仍不顯著, 結論一致

Tchexper 係改變小, 表示在控制學校固定效果後,

Tchexper 的影响变小。

White_asian 係改大幅增加, 表示在控制學校固定效果後, White_asian 的影响更大

Boy 和 Freelunch 的影响方向和顯著性不變, 結論一致

(v)
F test for individual effects

```
data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch  
F = 16.698, df1 = 78, df2 = 5681, p-value < 2.2e-16  
alternative hypothesis: significant effects
```

H_0 : 所有的学校固定效果皆相同 v.s

H_1 : 至少1个学校固定效果不同

Assume: $\alpha = 0.05$

因為 $p\text{-value} = 2.2 \times 10^{-16} < 0.05$, 所以 reject H_0 .

有證據支持學校間存在顯著的個別差異, 使用
fixed-effect model 較合適。

(d)

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	436.126774	2.064782	211.2217	< 2.2e-16 ***
small	6.458722	0.912548	7.0777	1.466e-12 ***
aide	0.992146	0.881159	1.1260	0.2602
tchexper	0.302679	0.070292	4.3060	1.662e-05 ***
boy	-5.512081	0.727639	-7.5753	3.583e-14 ***
white_asian	7.350477	1.431376	5.1353	2.818e-07 ***
freelunch	-14.584332	0.874676	-16.6740	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 6158000

Residual Sum of Squares: 4332100

R-Squared: 0.29655

Adj. R-Squared: 0.29582

Chisq: 493.205 on 6 DF, p-value: < 2.22e-16

Lagrange Multiplier Test - (Breusch-Pagan)

data: readscore ~ small + aide + tchexper + boy + white_asian + freelunch

chisq = 6677.4, df = 1, p-value < 2.2e-16

alternative hypothesis: significant effects

White_asian 和 Freelunch 是和學校效果相關的變數，
因為可以反映學區特性。

Tchexper 也可能是和學校效果相關的變數，因為較好的
學校可能較易吸引經驗豐富的老師。

H_0 : 不存在個體隨機效果 ($\sigma_u^2 = 0$)

Assume: $\alpha = 0.05$

i) $p\text{-value} = 2.2 \times 10^{-16} < 0.05 \quad \therefore \text{reject } H_0$

有證據支持 $\sigma_u^2 \neq 0$ ，使用 fixed-effect model 或
random effect model 較適合。

1e)

```
[1] "--- 根據公式 (15.36) 的個別變數 t 檢定結果 ---"
> print(results)
  Variable    b_FE    b_RE t_statistic p_value Significant_5pct Var_FE
1      small  6.4902  6.4587    1.1460  0.2518              No 0.833500
2        aide  0.9961  0.9921    0.1284  0.8978              No 0.777400
3      tchexper  0.2856  0.3027   -1.9377  0.0527              No 0.005019
4 white_asian  8.0280  7.3505    1.2181  0.2232              No 2.358000
5    freelunch -14.5936 -14.5843   -0.0956  0.9239              No 0.774400
6         boy  -5.4559 -5.5121          NA          NA No (Var Diff < 0) 0.529400
  Var_RE
1 0.832700
2 0.776400
3 0.004941
4 2.049000
5 0.765100
6 0.529500
```

```
[1] "--- 標準 Hausman 檢定結果 (整體模型) ---"
> print(hausman_test_result)

Hausman Test

data:  readscore ~ small + aide + tchexper + boy + white_asian + freelunch
chisq = 13.809, df = 6, p-value = 0.03184
alternative hypothesis: one model is inconsistent
```

$$H_0: \beta_{FE, K} = \beta_{RE, K}$$

在 $\alpha = 0.05$ 下, small, aide, tchexper, white-asian, freelunch 皆不顯著, 無法拒絕 H_0 . 沒有證據支持 FE 和 RE 的係數有差異, 表示二者估計值接近

因為 boy 的 $Var_{FE} - Var_{RE} < 0$, 所以無法使用 15.36 計算

(f)

Coefficients:

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	461.141562	19.562399	23.5729	< 2.2e-16 ***
small	6.489608	0.912846	7.1092	1.167e-12 ***
aide	0.999769	0.881559	1.1341	0.25676
tchexper	0.285536	0.070836	4.0310	5.555e-05 ***
white_asian	8.029241	1.535462	5.2292	1.702e-07 ***
freelunch	-14.593369	0.879896	-16.5853	< 2.2e-16 ***
boy	-5.456393	0.727498	-7.5002	6.371e-14 ***
mean_small	-19.359113	21.354865	-0.9065	0.36465
mean_aide	14.247074	19.546495	0.7289	0.46607
mean_tchexper	0.965602	0.596319	1.6193	0.10539
mean_white_asian	-7.001342	6.038434	-1.1595	0.24627
mean_freelunch	-3.942906	8.393178	-0.4698	0.63852
mean_boy	-52.424247	24.199967	-2.1663	0.03029 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Linear hypothesis test:

```
mean_small = 0
mean_aide = 0
mean_tchexper = 0
mean_white_asian = 0
mean_freelunch = 0
mean_boy = 0
```

Model 1: restricted model

Model 2: readscore ~ small + aide + tchexper + white_asian + freelunch +
boy + mean_small + mean_aide + mean_tchexper + mean_white_asian +
mean_freelunch + mean_boy

	Res.Df	Df	Chisq	Pr(>Chisq)
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1	5759			
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2	5753	6	13.38	0.03738 *
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Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$$H_0: r_{\text{small}} = r_{\text{aide}} = r_{\text{tchexper}} = r_{\text{white-asian}} = r_{\text{freelunch}} = r_{\text{boy}} = 0$$

$$) \text{ } p\text{-value} = 0.03738 < \alpha = 0.05$$

∴ reject H_0 . 有證據支持學校層級的平均值聯合起來對
readscore 有解釋力。